

Engineering report template

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Description

This document intends to document blah blah blah

Standard packages

```
using Pkg
# Ensure packages are available in the current environment
# Pkg.add(["DataFrames", "CSV", "Plots", "ControlSystems", "LinearAlgebra", "LaTeXStrings"])

using LinearAlgebra, Statistics
using DataFrames, CSV
using Plots
plotly()
using PlotThemes
# Detect if we are rendering in a headless/build environment like Quarto
const IS_RENDERING = haskey(ENV, "QUARTO_PROJECT_ROOT") ||
                    haskey(ENV, "QUARTO_DOCUMENT_PATH") ||
                    !isinteractive()

if IS_RENDERING
    # --- Light/High-Contrast Settings for Quarto HTML ---
    theme(:default) # Swaps to a clean, light-friendly palette
    default(
        background_color      = :white,
        background_color_inside = :white,
        background_color_legend = RGBA(0, 0, 0, 0.1), # Keep a slight dark tint for legend
        readability
        foreground_color_legend = nothing,

        # --- Grid & Frame Tuning ---
        grid      = true,
        gridalpha = 0.12, # Keep grid lines faint so they sit nicely over your
        theme
        linewidth = 1.3,

        # --- Layout & Geometry ---
        size      = (600, 300), # Good aspect ratio for a horizontal 1x2 layout
        legend     = :best
    )
else
    theme(:dark)
    default(
        # --- The Transparency Magic ---
        background_color      = :transparent, # Makes the main plot area transparent
        background_color_inside = :transparent, # Makes the inside of the axes transparent
        background_color_legend = RGBA(0, 0, 0, 0.4), # Keep a slight dark tint for legend
        readability
```

```

foreground_color_legend = nothing,

# --- Grid & Frame Tuning ---
grid      = true,
gridalpha = 0.12,      # Keep grid lines faint so they sit nicely over your
                        theme
linewidth = 1.3,

# --- Layout & Geometry ---
size      = (600, 300), # Good aspect ratio for a horizontal 1x2 layout
legend    = :best
)
end

using LaTeXStrings # Essential for math in labels
using Handcalcs, Unitful, UnitfulLatexify
using Format

# Handcalcs config
set_handcalcs(precision=2)
left_align_in_pluto() # I think this works with vscode?
reset_handcalcs()

using Latexify

using Revise
println("Julia Engineering Stack Loaded.")

```

Julia Engineering Stack Loaded.

Topic 1

Insert text for equation here. You can number and reference equation with [Equation 1](#)

$$x = a_2 \tag{1}$$

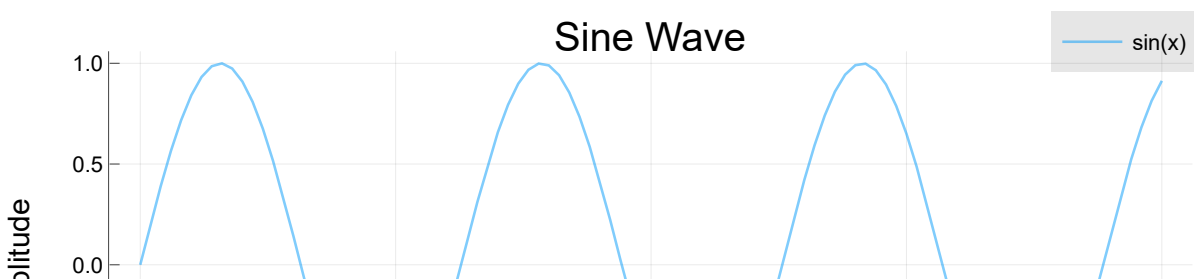
Sample plot code below

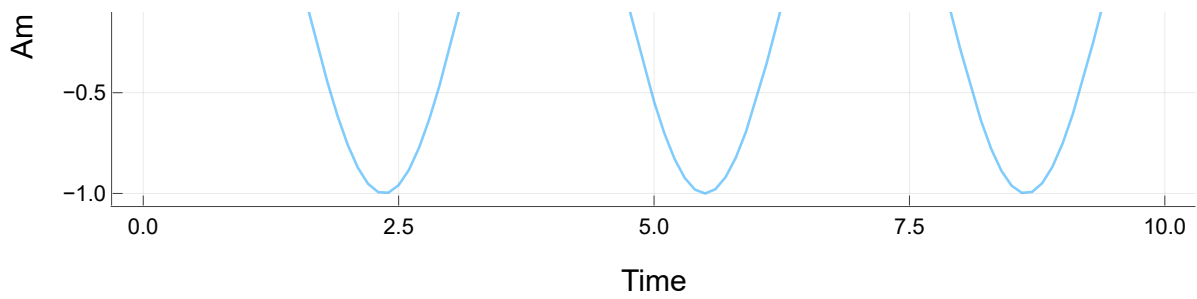
```

x = 0:0.1:10
y = sin.(2*x)

# Create a line plot
plot(x, y, title="Sine Wave", xlabel="Time", ylabel="Amplitude", label="sin(x)", sp=1,
      alpha=0.5)

```





```
# Here is an example of a Handcalcs
I_1 = 10.5123123u"mA"
R_1 = 1u"Ω"
@handcalcs begin
    P_1 = R_1 * I_1^2 |> u"mW"
end precision=2
```

$$P_1 = R_1 \cdot I_1^2 = 1 \, \Omega \cdot (10.51 \, \text{mA})^2 = 0.11 \, \text{mW}$$